

Explainable Graph-Based Spatial Vulnerability Mapping for Glaucoma Using Fundus Images

S. Ravi Kishan
Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha
Engineering College
Vijayawada, India
suraki@vrsiddhartha.ac.in

Sudikonda Sri Srinivasa Sathwik
Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha
Engineering College
Vijayawada, India
ssathwik31831@gmail.com

Sridhara Taraka Sri Balaji
Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha
Engineering College
Vijayawada, India
tarakasribalajisridhara@gmail.com

Abstract—The majority of current methods that describe how glaucoma evolves over time rely on expensive 3D Optical Coherence Tomography (OCT) scans or lengthy Visual Field (VF) test sequences, which restricts their scalability for population-level screening. Conversely, 2D retinal fundus photography is readily accessible, but standard convolutional neural networks (CNNs) function as static, black-box detectors that simply classify an eye as glaucomatous or healthy, failing to identify the areas of the optic nerve that might be most structurally vulnerable. Instead of trying to perform full temporal prognosis from a single image, this paper presents a graph-theoretic framework that transforms Grad-CAM outputs from a dual-specialist CNN into spatial risk pathways rooted at the optic cup, providing interpretable maps of where the tissue appears most susceptible. In order to ensure localized and anatomically significant activations, a Disc-Peripheral dual-specialist CNN trained on the public ORIGA and REFUGE datasets first produces a Grad-CAM-based vulnerability heatmap. A "resistance network" that promotes traversal through high-risk tissue is produced by tessellating the fundus crop surrounding the optic nerve head into a superpixel Region Adjacency Graph, where edge resistances are defined as an inverse function of Grad-CAM risk. Spatial risk trajectories that identify structurally vulnerable areas are produced by using Dijkstra's shortest-path algorithm from the optic cup to automatically chosen high-risk targets. Although the model was never explicitly encoded with the clinical ISNT rule, 61.8% of these trajectories on 1,050 fundus pictures lie inside the inferior and superior quadrants, reflecting the biomechanically vulnerable poles defined by the rule. These findings imply that graph-based, explainable spatial risk maps that are biologically consistent can supplement single-image fundus screening and help clinicians concentrate on structurally vulnerable areas. However, longitudinal data will be needed in subsequent research to confirm true prognostic value.

Index Terms—Explainable AI, Spatial Risk Mapping, Grad-CAM, ISNT Rule, Graph-Theoretic Modeling, Glaucoma Screening.

I. INTRODUCTION

A. Background and Motivation

Because glaucoma damages the optic nerve, it can result in irreparable vision loss if it is not caught early. Nowadays, artificial intelligence (AI) models are mostly employed to analyze eye scans and assess a patient's health or condition [1]. Clinicians frequently need more detailed information than a straightforward "glaucoma" or "no glaucoma" designation,

even though such binary classification is helpful. They would particularly benefit from spatial indications regarding the areas of the peripapillary retina and optic nerve head (ONH) that seem to be most structurally fragile, since this could direct more careful monitoring and treatment choices.

However, many extremely accurate AI models for glaucoma function as "black boxes," making predictions without providing a clear explanation of how or why. Clinicians find it challenging to trust AI systems in crucial medical workflows because of this opacity, particularly when management choices rely on comprehending the anatomical foundation of model outputs [6]. The development of AI technologies that not only identify glaucoma but also generate transparent, anatomically based visualizations of spatial vulnerability that physicians can examine and explain is therefore highly motivated.

B. Problem Definition

The majority of current AI algorithms for glaucoma from fundus photos can only identify the disease's current condition. They do not indicate which areas seem more vulnerable to additional injury based on baseline structure, nor do they give clear spatial information about how vulnerability is dispersed throughout the optic nerve and surrounding tissue. Furthermore, longitudinal data—that is, many examinations of the same patient over a number of years—is necessary for many approaches that try to describe progression patterns [2]. This is costly, time-consuming, and frequently unfeasible in environments with limited resources. Graph Convolutional Networks (GCNs) have been used in recent work to better represent the architecture of the eye, however these methods usually employ graphs to accomplish segmentation or to define static regions like the disc and cup [1], [7]. The propagation of AI-estimated vulnerability through tissue connection is not explicitly modeled by them. Building a transparent system that, from a single fundus image, generates *spatially explicit, anatomically constrained risk pathways* that highlight tissue regions appearing structurally susceptible is the challenge addressed in this work. It does not claim to accurately predict the precise future path of disease for a specific patient.

C. Research Gap

After studying the existing technologies, several gaps become apparent:

- **Lack of Single-Image Spatial Risk Mapping:** Three to five years of longitudinal data are used by many models that examine the evolution of glaucoma in order to describe structural or functional changes [2], [5]. There are still few techniques that can extract precise spatial risk patterns from a single baseline fundus imaging without the need for long-term monitoring.
- **Limited Integration of Explainability into Spatial Modeling:** Using advanced methods like generative models that create imaginary future visuals, several research try to predict future changes [3]. These, however, frequently lack straightforward, understandable structure-function relationships and might not offer biologically obvious, mathematically supported explanations for their localized predictions.
- **Static Use of Graph Networks:** Graphs are commonly used by current graph-based AI models for fundus pictures to either represent local spatial relationships or segment the eye into discrete parts [7]. To the best of our knowledge, not much research has been done to identify anatomically plausible risk pathways that originate from the optic cup using a geometric graph as a resistance network that is controlled by explainable AI outputs.

D. Contributions of This Work

In order to fill these gaps, this work introduces a novel, easily understood method that offers spatially precise risk information from a single fundus picture, going beyond simple illness identification. The following is a summary of the main contributions:

- **Single-Image Spatial Vulnerability Mapping:** Without the need for longitudinal VF or OCT data, we convert Grad-CAM activations from a glaucoma classifier into anatomically limited spatial risk trajectories that highlight tissue regions that appear structurally susceptible using a single baseline fundus image.
- **Active XAI-Based Resistance Network:** We use explainable AI as a quantitative indicator as well as a visual overlay. In order to build a resistance network that encodes tissue vulnerability in its path costs, we specifically divide the ONH region into a superpixel graph and insert Grad-CAM risk values into the edges.
- **Biologically Consistent Risk Pathways:** We create spatial risk trajectories whose destinations exhibit 61.8% alignment with the inferior and superior quadrants of the ISNT rule by using Dijkstra's shortest-path algorithm from the optic cup to autonomously selected high-risk objects. Although the rule is never explicitly encoded, this alignment implies that the suggested graph format recovers established patterns of structural susceptibility from fundus pictures alone.

II. LITERATURE SURVEY

Recent advances in medical image processing have explored the use of Graph Convolutional Networks (GCNs) to understand the complex physical structure of the eye. For instance, Lenka et al. [1] introduced a multi-task deep learning model that uses graph convolutions to improve glaucoma detection and segmentation from retinal fundus pictures. In a similar vein, Fu et al. [7] successfully illustrated the geometric boundaries of the optic disc and cup using graph theory and deep networks. These methods are mainly limited to static assessments that map the spatial relationships between different eye regions at one particular moment in time. The graphs do not simulate how vulnerability might spread through tissue connectivity; instead, they are used to depict current anatomy or establish region boundaries.

To address the need to understand how glaucoma spreads, researchers have also developed models that follow the disease's progression across time. Majid et al. [2] proposed a deep learning method to map the spatial distribution of glaucoma and identify early biomarkers by examining a set of longitudinal photographs. By analyzing serial optic disc photos, other techniques, like the weighted ensemble method by Liu et al. [5] and the deep learning model by Mohammadzadeh et al. [4], have been successful in predicting visual field deterioration and identifying rapid disease progressors. However, these methods are not appropriate for single-visit screening situations since they rely solely on longitudinal data and require a lengthy history of patient visits and historical records to estimate progression.

In order to forecast future structural changes without having to wait several years, generative algorithms have been developed. In order to predict vision loss prior to standard clinical testing, Hussain et al. [3] developed a multimodal architecture that uses a Generative Adversarial Network (GAN) to produce future images of the eye. Despite being new, this approach requires intricate multimodal datasets that combine VF testing and clinical data, which are rarely accessible in standard screening environments. Furthermore, when solely generative approaches "hallucinate" future visions, they offer no mathematical transparency, which could make physicians less confident in the localized forecasts.

Beyond structural imaging, the Visual Field (VF) test is the clinical gold standard for evaluating functional vision loss. Recent efforts to estimate and predict VF degradation have made use of deep learning. In their study of the application of deep learning in glaucoma, Thompson et al. [8] pointed out that while structural changes often precede functional loss, it is still difficult to anticipate precise VF impairments due to noise and variability in patient testing. In order to address this, new models have demonstrated that serial optic disc photos can be used to directly forecast functional VF progression [10], and multimodal architectures integrating OCT scans and prior VF data have been suggested to predict future VF maps [9]. However, these functional prediction systems still rely on longitudinal data and are not designed for settings where only

a single baseline fundus image is available.

III. METHODOLOGY

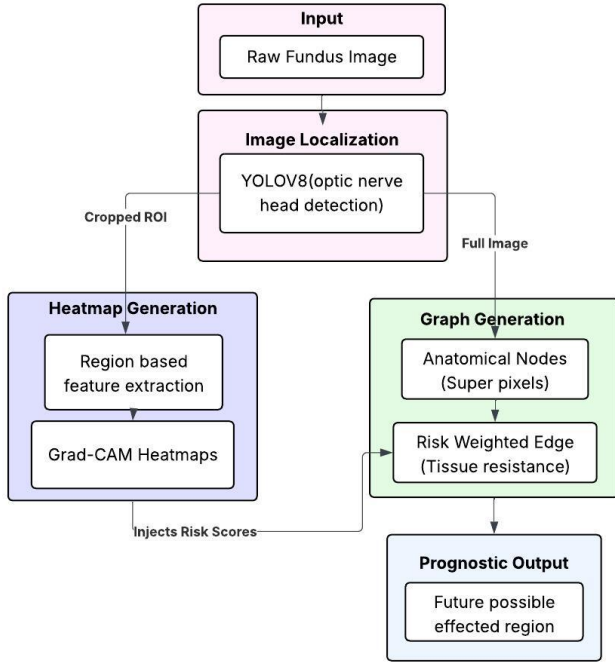


Fig. 1. Architecture diagram of the proposed spatial vulnerability mapping framework.

A. Dataset Description and Preparation

This study uses a combined dataset of 1,050 retinal fundus images to validate the graph-based spatial risk framework and train the underlying deep learning models. These images were gathered from two publicly accessible databases:

- **REFUGE:** 400 images (360 healthy, 40 glaucoma).
- **ORIGA:** 650 images (482 healthy, 168 glaucoma).

Combining the datasets yields 1,050 total images (842 healthy, 208 glaucomatous), resulting in a natural 4:1 class imbalance that is typical of real-world medical data. To robustly train the classifier despite this skew, we utilized:

- **Stratified Sampling:** guarantees that the 4:1 ratio is rigorously upheld during training and testing splits, avoiding data leakage and offering a representative assessment.
- **Class Weighting:** gives the minority glaucoma class a larger loss penalty. Instead of just forecasting the majority healthy class, this drives the algorithm to acquire characteristics linked to ill eyes.

B. Data Preprocessing

Before being converted into a graph, each fundus image needs to be localized and standardized. The raw fundus pictures are first subjected to Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE aids the model in

detecting minute tissue flaws by improving the visual contrast and clarity of the vascular network and neuroretinal structures.

Next, manual cropping is replaced with a custom YOLOv8 object detection module. A tight bounding box is created around the Optic Nerve Head (ONH), which is automatically identified. This ensures that further investigation concentrates on the anatomically important area by isolating the optic nerve head and peripapillary region.

C. Dual-Specialist Architecture and Merged Risk Mapping

A pre-trained deep learning algorithm examines the optic nerve region following localization. To guarantee that both central and peripheral structures are sufficiently captured, a "Dual-Specialist" architecture is employed. The *Peripheral Specialist* employs a donut-shaped mask to examine surrounding retinal tissue, while the *Disc Specialist* concentrates on the optic nerve head center.

Both specialized networks use Grad-CAM, a XAI approach, to produce class-activation heatmaps. The spatial regions that most significantly affect the glaucoma prediction are revealed by Grad-CAM. After that, a single *Merged Risk Heatmap* is created by combining the two heatmaps. High Grad-CAM ratings are assigned to pixels that represent tissue that looks weak, damaged, or extremely suspicious, whereas lower scores are assigned to healthier areas.

D. Graph Construction (The Resistance Network)

Creating a structural mesh from the 2D risk map is the fundamental mathematical step. The Simple Linear Iterative Clustering (SLIC) technique divides the ONH image into hundreds of connected segments, called *superpixels*. Every superpixel functions as a node in the graph and represents a tiny tissue area.

Next, nearby superpixels are connected to create a Region Adjacency Graph (RAG). The vulnerability scores obtained from Grad-CAM are inserted into the graph's edges. To simulate physical resistance, we use a straightforward mapping:

$$\text{Resistance} = \max(1.0 - \text{Average Risk}, 0.05) \quad (1)$$

where *Average Risk* is the average Grad-CAM score of two nearby nodes. Zero-resistance edges, which could lead to unstable routing algorithms, are prevented with a floor of 0.05. The static image is successfully converted into a dynamic "Resistance Network" by this mapping, where pathways through high-risk tissue have reduced resistance.

E. Spatial Constraints and Target Selection

The destinations need to be specified before calculating spatial risk pathways. Because these nodes frequently cluster within a single lesion, creating duplicate routes, it would be clinically useless to just select the top three highest-risk nodes.

A dual-constraint target selection technique is employed to prevent this:

- **Source Exclusion Rule:** The model is encouraged to search beyond already severely injured tissue by excluding nodes within a radius of 60 pixels from the optic cup center as targets.

- **Mutual Separation Rule:** Average risk is used to rank the remaining nodes. In order to promote spatially different regions, a candidate target is only approved if its centroid is at least 80 pixels apart from every target that has already been chosen.

These constraints reduce clustering and produce three distinct, high-risk retinal regions that serve as endpoints for path computation.

F. Spatial Risk Routing via Shortest-Path Algorithm

The center of the optic cup, which represents tissue that has already experienced significant damage, is chosen as the source node in order to calculate spatial risk trajectories. The resistance network is next subjected to Dijkstra's shortest-path algorithm.

The shortest pathways preferentially pass across high-risk superpixels while maintaining anatomical connectedness since edge resistances are negatively correlated with susceptibility. By following these routes from the source to each of the three chosen destinations, one can create *Predictive Risk Maps*, which show likely spatial trajectories of susceptibility instead of making firm predictions about future damage.

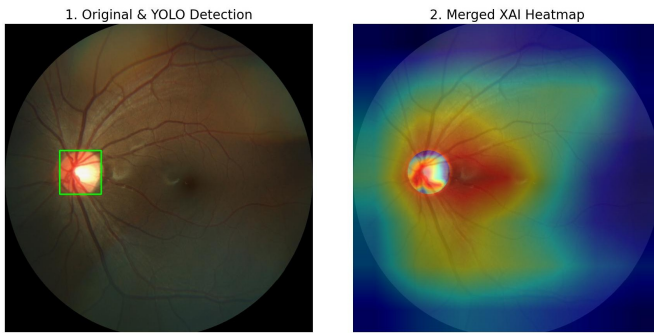


Fig. 2. Examples of optic disc localization and corresponding Grad-CAM risk heatmaps.

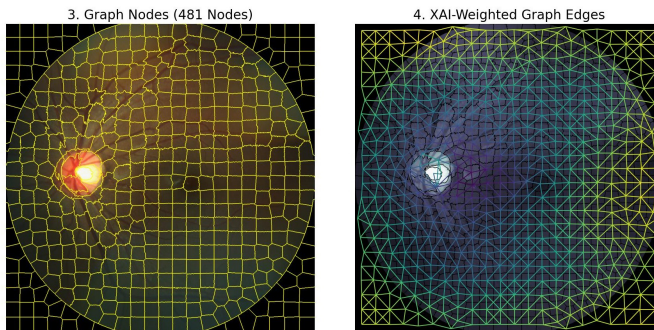


Fig. 3. Retinal region segmented into superpixel nodes and overlaid with the triangulated resistance network mesh.

G. Geometric ISNT Classification

An automatic geometric classifier is presented to connect computational trajectories to anatomical terms. The system

calculates the trajectory vector from the source (optic cup center) to each target rather than using the target nodes' precise locations.

Let the centroid of a predicted target node be (t_x, t_y) , and the center of the optic cup be (c_x, c_y) . The trajectory angle is given by

$$\theta = \arctan 2(t_y - c_y, t_x - c_x) \times \frac{180}{\pi} \quad (2)$$

In standard image coordinates, positive vertical direction corresponds to increasing y values, so a positive angle indicates a downward trend. The angles are mapped to anatomical poles as follows:

- **Inferior (Bottom) Pole:** $45^\circ \leq \theta \leq 135^\circ$
- **Superior (Top) Pole:** $-135^\circ \leq \theta \leq -45^\circ$
- **Nasal/Temporal (Horizontal) Poles:** All remaining angles.

This trigonometric classification makes the evaluation against the ISNT rule objective and reproducible.

H. Implementation Details and Hyperparameters

Python was used to implement the system. Deep learning was done with PyTorch, while image processing and graph creation were done with Scikit-Image and NetworkX, respectively. Important implementation specifics include:

- **Image Standardization:** All ONH crops were resized to 1024×1024 pixels. CLAHE was applied in the LAB color space with a clipLimit of 2.0 and tileGridSize of (8, 8).
- **Localization Restrictions:** To avoid incorrectly treating the entire eyeball as the optic nerve, YOLOv8 detections were accepted only if the bounding box area was less than 25% of the full image.
- **Feature Extraction and XAI:** Both Disc and Peripheral specialists are based on ResNet-18, and Grad-CAM is computed from the final convolutional layer (layer4[-1]). Combined pixel risk scores are normalized to the range [0.0, 1.0].
- **Graph Tessellation Parameters:** The SLIC algorithm is configured with $n_segments = 500$ and compactness = 15 to generate superpixels with approximately homogeneous size and smooth boundaries.
- **Edge Weight Formulation:** Edge resistances are computed using (1), with a minimum floor of 0.05 to ensure numerical stability.
- **Routing Thresholds:** The three highest-risk, spatially distinct targets are selected, with a minimum distance of 60 pixels from the source and mutual separation of at least 80 pixels.

IV. RESULTS AND DISCUSSION

A. Biological Consistency via the ISNT Rule

The ISNT rule is used as a structural reference to evaluate the biological plausibility of the spatial risk trajectories. In normal eyes, early glaucomatous damage frequently occurs in

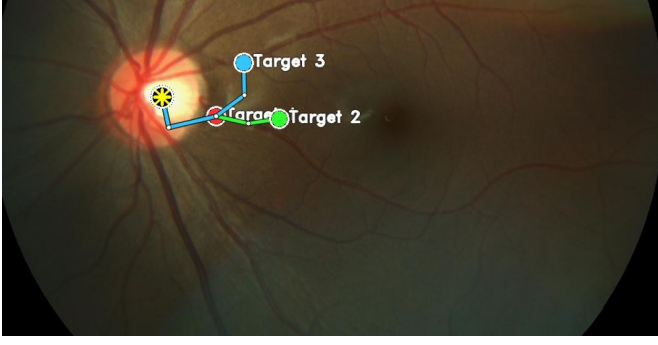


Fig. 4. Example of a retinal image showing the source at the optic disc and computed spatial risk paths to three selected high-risk targets.

the inferior and superior poles of the neuroretinal rim, which are known to be biomechanically sensitive. [11] [12]

All 1,050 photos in the combined REFUGE and ORIGA dataset were subjected to shortest-path routing. Three spatial risk trajectories were created for every image, yielding 3,150 routes. One of three anatomical categories—superior, inferior, or nasal/temporal—was assigned to each trajectory.

B. Distribution of Spatial Risk Trajectories

The estimated trajectories and known anatomical susceptibility exhibited a substantial correlation, according to the analysis. 971 (30.8%) and 977 (31.0%) of the 3,150 pathways ended in the inferior and superior quadrants, respectively. The inferior and superior poles, which are precisely the areas predicted to be more vulnerable according to the ISNT rule, accounted for 61.8

Although neither the ISNT rule nor explicit quadrant limits were hardcoded into the graph or routing algorithm, this alignment still appears. Rather, the paths of least resistance, as determined by anatomical connection and Grad-CAM risk levels, give rise to the trajectories. These results imply that the percentage of paths aimed at the vertical poles falls within the general range of ISNT-related performance reported in population-based disc-photograph studies [13], and that the single-image resistance network captures clinically significant patterns of structural susceptibility.

TABLE I
COMPARISON OF DIFFERENT MODEL ARCHITECTURES USED IN GENERATING GRAD-CAM FROM FUNDUS IMAGES.

Model	Methodology	Performance
ResNet-18	Visual CNN only	Accuracy: 0.79 AUC: 0.8341
YOLOv8 only	VCDR-based detection	Accuracy: 0.91 AUC: 0.86
Proposed Hybrid	CNN + YOLO fusion	Accuracy: 0.86 AUC: 0.93

As shown in Table I, the proposed Hybrid framework achieves an AUC of 0.93, outperforming standalone ResNet-18 (0.8341) and YOLOv8 (0.86). Although YOLOv8 yields the highest raw accuracy (0.91), accuracy can be misleading in the

Cross-Dataset Validation: Forecasted Pathways vs. Anatomical ISNT Rule
(Tested on 1050 Images / 3150 Forecasted Routes)

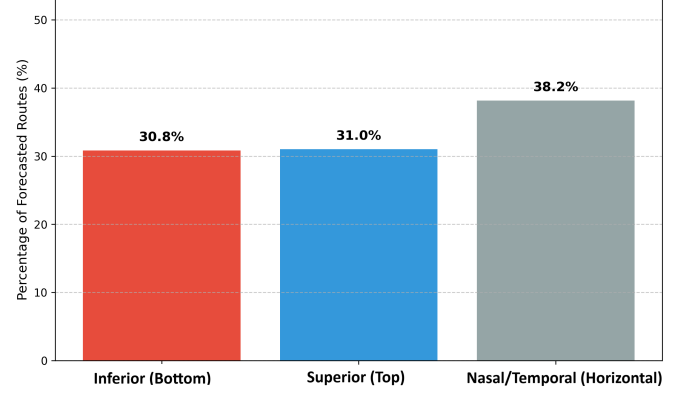


Fig. 5. Cross-dataset validation results, showing the proportion of spatial risk trajectories terminating in each anatomical quadrant and their alignment with the ISNT rule.

presence of a 4:1 class imbalance, as a model can achieve high accuracy by over-predicting the majority class. The Hybrid model's higher AUC indicates better discrimination between healthy and glaucomatous eyes, providing a more reliable foundation for Grad-CAM-based risk mapping.

TABLE II
ISNT-RELATED METRICS FROM PRIOR WORK COMPARED TO THE PROPOSED MODEL.

Study	ISNT-related Metric	Reported Value
Harizman et al. (2006) [12]	Odds ratio for glaucoma when ISNT rule violated	6.04
Poon et al. (2017) [14]	Eyes strictly obeying ISNT rule	37.0%
Poon et al. (2017) [14]	Eyes obeying IS / IST variants	70.9–76.4%
Maupin et al. (2020) [13]	ISNT sensitivity / specificity	94.1% / 49.2%
Maupin et al. (2020) [13]	ISNT sensitivity / specificity	79.4% / 34.1%
Proposed model	Paths ending in inferior or superior quadrants	61.8%

C. ISNT-based state-of-the-art and external validation

Because this work operates on single fundus images without longitudinal ground truth, the ISNT rule provides a natural external anatomical reference for validation. The rule states that in normal eyes the neuroretinal rim is typically thickest inferiorly, followed by superior, nasal, and temporal sectors ($I \geq S \geq N \geq T$), and that deviations from this pattern are associated with glaucomatous damage [11], [12]. In stereo disc photographs, Harizman et al. reported that violation of the ISNT rule is strongly associated with glaucoma, with an odds ratio of 6.04 [12]. Poon et al. then examined normal eyes and found that only 37.0% of disc photo rim assessments and 43.8% of RNFL measurements strictly obeyed the ISNT pattern, whereas IST and IS variants were valid in 70.9–76.4% of disc photos and 70.9–71.8% of RNFL measurements [14]. Maupin et al. quantified diagnostic performance in a population-based cohort, reporting that for monoscopic disc

photographs the ISNT rule achieved 94.1% sensitivity and 49.2% specificity, while for SD-OCT RNFL thickness the sensitivity and specificity were 79.4% and 34.1%, respectively [13].

In contrast to these studies, which evaluate how well the ISNT pattern itself separates normal from glaucomatous eyes in disc photos or SD-OCT scans, the present work uses the ISNT rule as an external consistency check for graph-based spatial risk trajectories derived from 2D fundus images. Across 1,050 fundus photographs, 61.8% of all computed spatial risk paths terminated in the inferior or superior quadrants, the poles that ISNT-based analyses identify as structurally vulnerable and commonly affected in early glaucoma. Although this path-level alignment is not directly comparable to patient-level sensitivity or specificity, it is consistent with ISNT-based analyses that highlight the inferior and superior poles as structurally vulnerable in early glaucoma [13], [14].

V. CONCLUSION AND FUTURE WORK

This work introduces a graph-theoretic approach that uses transparent, spatially explicit vulnerability mapping to enhance static glaucoma detection from fundus pictures. Grad-CAM scores from a dual-specialist CNN are incorporated into a superpixel-based resistance network using a single baseline fundus image, and shortest-path routing from the optic cup produces spatial risk trajectories that identify structurally vulnerable areas. The suggested approach recovers biologically realistic patterns of structural susceptibility from fundus pictures alone, as evidenced by the observed 61.8% alignment of trajectory destinations with the inferior and superior quadrants of the ISNT rule.

Crucially, rather than being a completely validated temporal prognosis, the current study should be seen as *spatial risk mapping*. The trajectories do not indicate future outcomes for specific individuals; rather, they show likely anatomical routes of vulnerability conditioned on baseline structure. In this single-visit context, direct empirical comparison with longitudinal progression models is not yet feasible.

Three directions will be the focus of future effort. First, a quantitative evaluation of how well the spatial risk trajectories match actual future structural and functional changes will be made possible by the integration of longitudinal clinical information. Second, in order to assess spatial pathway predictions, new standardized metrics that go beyond conventional scalar accuracy measures are required. Third, tests involving clinicians in the loop will be carried out to determine whether graph-based risk maps, as opposed to traditional Grad-CAM overlays, enhance interpretability and decision-making.

This work offers a fundamental step toward more transparent, geographically informed glaucoma screening from widely accessible fundus pictures by clearly structuring the procedure as an explainable spatial vulnerability tool and confirming its results against established anatomical criteria.

REFERENCES

- [1] S. Lenka, Z. L. Mayaluri, G. Panda, and S. R. Nayak, "Glaucoma detection from retinal fundus images using graph convolution based multi-task model," *e-Prime – Advances in Electrical Engineering, Electronics and Energy*, vol. 11, art. no. 100931, 2025.
- [2] I. Majid, Z. Mishra, Z. C. Wang, V. Chopra, D. Heuer, and Z. J. Hu, "Automated detection and biomarker identification associated with the structural and functional progression of glaucoma on longitudinal color fundus images," *Applied Sciences*, vol. 15, p. 1627, 2025, doi: 10.3390/app15031627.
- [3] S. Hussain, J. Chua, D. Wong, J. Lo, A. Kadziauskiene, R. Asoklis, G. Barbastathis, L. Schmetterer, and L. Yong, "Predicting glaucoma progression using deep learning framework guided by generative algorithm," *Scientific Reports*, vol. 13, art. no. 19960, 2023.
- [4] V. Mohammadzadeh, S. Wu, T. Davis, A. Vepa, E. Morales, S. Besharati, K. Edalati, J. Martinyan, M. Rafiee, A. Martynian, F. Scalzo, J. Caprioli, and K. Nouri-Mahdavi, "Prediction of visual field progression with serial optic disc photographs using deep learning," *BMJ Open Ophthalmology*, vol. 9, no. 1, 2024.
- [5] Y. Liu, X. Chen, H. Li, and J. Zhang, "Predicting glaucoma visual field progression from fundus images with a weighted ensemble method," in *Proc. IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2024.
- [6] S. Hussain, J. Chua, D. Wong, L. Schmetterer, and L. Yong, "Explainable artificial intelligence (XAI) in glaucoma diagnosis," *Scientific Reports*, vol. 13, 2023.
- [7] H. Fu, J. Cheng, Y. Xu, D. W. K. Wong, J. Liu, and X. Cao, "Joint optic disc and cup segmentation based on multi-label deep network and polar transformation," *IEEE Transactions on Medical Imaging*, vol. 38, no. 1, pp. 159–171, Jan. 2019.
- [8] A. C. Thompson, A. A. Jammal, and F. A. Medeiros, "A review of deep learning for screening, diagnosis, and detection of glaucoma progression," *Translational Vision Science & Technology*, vol. 9, no. 2, 2020.
- [9] Q. Pham Tran Minh, J. C. Han, D. Y. Park, and J. Shin, "Multimodal deep learning model of predicting future visual field for glaucoma patients," *IEEE Access*, early access, pp. 1–1, 2023, doi: 10.1109/ACCESS.2023.3248065.
- [10] F. Li, Y. Su, F. Lin, Z. Li, Y. Song, S. Nie, J. Xu, L. Chen, S. Chen, H. Li, K. Xue, H. Che, Z. Chen, B. Yang, H. Zhang, M. Ge, W. Zhong, C. Yang, L. Chen, F. Wang, Y. Jia, W. Li, Y. Wu, Y. Li, Y. Gao, Y. Zhou, K. Zhang, and X. Zhang, "A deep-learning system predicts glaucoma incidence and progression using retinal photographs," *The Journal of Clinical Investigation*, vol. 132, no. 11, art. no. e157968, 2022, doi: 10.1172/JCI157968.
- [11] J. B. Jonas, G. C. Gusek, and G. O. H. Naumann, "Optic disc, cup and neuroretinal rim size, configuration and correlations in normal eyes," *Investigative Ophthalmology & Visual Science*, vol. 29, no. 7, pp. 1151–1158, 1988.
- [12] N. Harizman, C. Oliveira, A. Chiang, C. Tello, M. Marmor, R. Ritch, and J. M. Liebmann, "The ISNT rule and differentiation of normal from glaucomatous eyes," *Archives of Ophthalmology*, vol. 124, no. 11, pp. 1579–1583, 2006.
- [13] E. Maupin, F. Baudin, L. Arnould, A. Seydou, C. Binquet, A. M. Bron, and C. P. Cruzot-Garcher, "Accuracy of the ISNT rule and its variants for differentiating glaucomatous from normal eyes in a population-based study," *British Journal of Ophthalmology*, vol. 104, no. 10, pp. 1412–1417, 2020, doi: 10.1136/bjophthalmol-2019-315554.
- [14] L. Y. C. Poon, D. Solá-Del Valle, A. Turalba, J. L. Klinenberg, L. R. Pasquale, and F. A. Medeiros, "The ISNT rule: How often does it apply to disc photographs and retinal nerve fiber layer measurements in the normal population?," *American Journal of Ophthalmology*, vol. 184, pp. 19–27, 2017.

- [1] S. Lenka, Z. L. Mayaluri, G. Panda, and S. R. Nayak, "Glaucoma detection from retinal fundus images using graph convolution based multi-task